AWNON BHOWMIK

CONTACT INFORMATION

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RESEARCH INTERESTS

Number Theory	Cryptography	Fractal Geometry
Computational Mathematics	Mathematical Modeling	Algorithm Design & Analysis

EDUCATION

Degree	**Institution**	**Concentration**	**Obtained**
Ph.D.	University of Central Florida	Mathematics	Expected 2026
MS	The City College of New York	Mathematics	December 2021
BS	CUNY York College	Major: Mathematics Minor: Computer Science	January 2020
AS	CUNY BMCC	Mathematics	August 2015

PUBLICATIONS

[1] **A. Bhowmik** and U. Menon, "An adaptive cryptosystem on a finite field," *PeerJ Computer Science*, vol. 7, e637, Aug. 2021. DOI: https://doi.org/10.7717/peerj-cs.637.

[2] **A. Bhowmik**, "An encoding schematic based on coordinate transformations," *International Journal of Mathematical Sciences and Computing (IJMSC)*, vol. 6, no. 6, pp. 9–14, Dec. 2020, ISSN: 2310-9033. DOI: 10.5815/ijmsc.2020.06.02. [Online]. Available: http://www.mecs-press.org/ijmsc/ijmsc-v6-n6/IJMSC-V6-N6-2.pdf.

[3] **A. Bhowmik** and U. Menon, "Dragon crypto – an innovative cryptosystem," *International Journal of Computer Applications (IJCA)*, vol. 176, no. 29, pp. 37–41, Jun. 2020, ISSN: 0975-8887. DOI: 10.5120/ijca2020920331. [Online]. Available: http://www.ijcaonline.org/archives/volume176/number29/31386-2020920331.

[4] **A. Bhowmik** and U. Menon, "Enhancing the ntru cryptosystem," *International Journal of Computer Applications (IJCA)*, vol. 176, no. 29, pp. 46–53, Jun. 2020, ISSN: 0975-8887. DOI: 10.5120/ijca2020920320. [Online]. Available: http://www.ijcaonline.org/archives/volume176/number29/31388-2020920320.

[5] **A. Bhowmik** and U. Menon, "Mes – modern encryption standard," *International Journal of Computer Applications (IJCA)*, vol. 176, no. 36, pp. 21–27, Jul. 2020, ISSN: 0975-8887. DOI: 10.5120/ijca2020920479. [Online]. Available: http://www.ijcaonline.org/archives/volume176/number36/31435-2020920479.

RESEARCH EXPERIENCE

Analysis of Searching & Sorting Algorithms November 2019–December 2019

- Using files with a varying number of inputs, applied linear and binary search. Analyzed the execution time for different inputs and proved algebraically and graphically – the time complexity of those algorithms.
- Implemented various common sorting algorithms such as bubble sort, merge sort, etc. on files containing varying amounts of data. Analyzed the execution time for different inputs and proved algebraically and graphically – the time complexity of those algorithms.

Advisor: Dr. Louis D' Alotto
Department of Mathematics & Computer Science, CUNY York College

Battery Powered Car Problem September 2019

- Given a pile of n batteries assuming an ideal mileage of k kilometers per battery, derived a recursive function that models the effective distance traveled by car since it can only carry one spare battery in it along with the other that powers its engine.
- Came up with two different algorithms and implemented using C++ code, to demonstrate the validity of the mathematical perspective. Illustrated with graphs to further support the claims.

Advisor: Dr. Louis D' Alotto
Department of Mathematics & Computer Science, CUNY York College

CSTEP Research Project September 2014–August 2015
The Electrolytic Properties of Protein in the Cell Membrane Channel

- Studied the Gramicidin A ion channel of the bacillus Brevis bacteria cell, containing the major amino acid Tryptophan.
- Modeled and conjectured that the conformational change in the Tryptophan produces changes in the cell membrane, thus behaving like a capacitive dielectric thereby influencing the ion current.
- Applied classical physics to a biological problem to mathematically model the ion current under the influence of a variable membrane dielectric, as a function of amino acid conformation.

Advisor: Dr. Brett A. Sims
Department of Mathematics, CUNY BMCC

Research Project: Safe Distance November 2014−December 2014

- Modeled the relationship between the initial speed of two cars, the reaction time, and braking distance, relating to ”average car lengths” as the unit of measure, essentially proving the ”Two Second Rule” implemented by the New York City Department of Motor Vehicles (NY DMV).
- Used classical probability and graphs to determine that this rule may change during weathers that hinder visibility and that the distance between two cars must be greater in monsoon and winter than in summer − before the brakes are applied to avoid a collision.

Advisor: Dr. David Waldman
Department of Science, CUNY BMCC

WORK EXPERIENCE

Graduate Teaching Grader January 2022−Present

- Assisting a faculty member in the non-instructional aspects of course teaching.
- Provides assistance to the students as required by the instructor.
- Running Office Hours in Zoom.
- Grading Quiz Papers and Test Papers.
- Providing feedback to the students.
- Inputting grades into WebcoursesUCF after grading and answering queries from students.

Department of Mathematics, University of Central Florida

Graduate Teaching Assistant January 2022−Present

- Assists students in setting up their accounts with online mathematics platform such as ALEKS, WebAssign, etc.
- Provides assistance to the students as required by the instructor
- Provides students with reading and writing material such as pen and paper, if requested
- Be available to answer students' queries during Open Lab sessions
- Ocassionally provides students with short lectures if requested by the instructor.
- Suggests a variety of resources such as videos, books, and other materials
- Performs proctoring duties during exam or testing hours if needed
- Adheres to the policies set by the mathematics lab and enforce the students to follow suit

Mathematics Assistance & Learning Lab (MALL), University of Central Florida

Substitute Teacher October 2020−Present

- Perform instructional and classroom management processes for teachers for one or more days.
- Adhere to the curriculum and lesson plans assigned by the regular teacher.
- Oversee students outside of the classroom including in the hallways and cafeteria.
- Take attendance and document daily notes.

NYC Department of Education

Mathematics Tutor – ASAP Program Jan 2020−May 2020

- Tutored around 30 college students each semester in subjects of Algebra, Quantitative Reasoning, Calculus, Differential Equations, and Physics to help them understand concepts and solve problems.
- Served as a liaison between professors and students and provided the students with updates regarding their progress and areas of challenge.

Department of Mathematics, CUNY BMCC

Adjunct College Lab Technician September 2017−August 2019

- Teach Calculus I in accordance to the professor's syllabus. Assist students with setting up learning exercises with the use of the software Maple and other laboratory materials. Provide extensive visual understanding of the theory taught in class.
- Use current technology to enhance educational effectiveness, including Maple and other online resources such as Desmos. Evaluate students' progress in attaining educational goals and objectives.
- Frequently meet in person or communicate via email with course instructor to discuss students' instructional programs.
- Track student assignments, attendance and test scores and enter into online database to provide real-time progress monitoring.

Department of Mathematics, CUNY BMCC

Lead Supplemental Instructor January 2017−August 2019

- Attended all classes for the assigned class and took notes in preparation for upcoming supplemental instructor sessions.
- Facilitated student learning and helped students better understand concepts or applications of course content.
- Frequently meet in person or communicate via email with course instructor to discuss students' instructional programs.
- Supported students and assisted them in gaining effective study skills by utilize diverse learning strategies and maximizing their potential for their academic success.
- Provided bi-weekly small group sessions covering the course content.

- Prepared structured SI sessions to review and reinforce difficult material introduced in class. In addition to group study and discussion, sometimes it included administering practice quizzes or teaching organizational and study practices to enhance mystery of the coursework and subject matter.

Department of Mathematics, CUNY BMCC

Mathematics & Physics Tutor December 2015−January 2018

- Tutored students ranging from 5th grade to undergraduate level on various mathematics and science courses.
- Reviewed classroom or curricula topics and assignments.
- Assisted students with homework, projects, test preparation, papers, research and other academic tasks.
- Worked with students to help them understand key concepts, especially those learned in the classroom.

Varsity Tutors

BMCC Immersion Project (Summer Program) June 2014−June 2015

- Conducted classes of 10 − 20 students and instructed them on various courses such as
  - Basic Arithmetic and Algebra (MAT 012)
  - Mathematics Literacy (MAT 041)
  - Elementary Algebra (MAT 051)
- The program allowed students to redo the material of a course they didn't pass so that they can complete the course. The Immersion workshops also provide incoming students with an opportunity to enroll in one course to reduce or eliminate the number of basic skills courses they would be required to take in the fall and or spring semester.
- The program also gave continuing students who were close to passing a 'basic skills' course an opportunity to work with a tutor in a small group or one-on-one setting. Upon completion of the tutorial program, students can test to satisfy their basic skills requirement before the fall or spring semester.

Department of Mathematics, CUNY BMCC

Mathematics Tutor − Math Lab March 2014−August 2015

- Tutored around 30 college students each semester in subjects of Algebra, Quantitative Reasoning, Calculus, Differential Equations, and Physics to help them understand concepts and solve problems.
- Served as a liaison between professors and students and provided the students with updates regarding their progress and areas of challenge.

Department of Mathematics, CUNY BMCC

AFFILIATIONS

- American Mathematical Society
- Society of Industrial & Applied Mathematics
- Phi Theta Kappa

ACHIEVEMENTS

- Dean's List, Fall 2019 – CUNY York College
- Dean's List, Winter 2015, Spring 2015, Fall 2014 – CUNY BMCC

HONORS & AWARDS

- Ernesto Malave Merit Scholarship, University Student Senate (USS) of CUNY
- Guttman Transfer Scholarship, Stella and Charles Guttman Foundation
- Inaugural Inter – University Math Olympiad Champion, University of Dhaka, July 2011

BROADER IMPACTS

Mathematics Stack Exchange

- Contributed 50+ answers to the community.
- Used this platform to learn about numerous books and problem-solving techniques.

QUORA

- Contributed to about 3500 answers related to mathematics and computer science.
- Dubbed the Top Writer in Calculus, Integration, and many other topics, numerous times.
- Answers became part of Quora Digest, multiple times.
- Impacted people all around the world, garnering a combined leadership of 4000 followers.
- Praised by experts in mathematics.

TECHNICAL SKILLS

Programming Languages	C, C++, Python, FORTRAN, SQL, MASM x86
Mathematical Packages	Mathematica, Maple, MATLAB
Typesetting Software	LaTeX
Image Processing Software	Adobe Photoshop, GIMP, Inkscape
3D Modeling and Animation Package	Autodesk 3ds Max
GUI Framework	Qt C++, Tkinter